

Personal Finance Tracker - Test & Validation Report

This document summarizes unit testing, UI validation, backend API validation, database verification, and end-to-end use cases for the Personal Finance Tracker application.

1. Unit Tests

Test Case	Expected Result	Status
TransactionServiceTest	Create transaction successfully	PASS
BudgetServiceTest	Prevent duplicate monthly budget	PASS
DashboardServiceTest	Correct income/expense calculations	PASS

Figure 1 - Login UI

- Responsive Login Screen
- JWT Authentication
- Material UI Components

Figure 2 - Dashboard UI

- Income vs Expense Cards
- Pie Chart
- Budget vs Actual Table

Figure 3 - Transaction Management

- Add/Edit/Delete
- Search & Filter
- Responsive Table

Figure 4 - Budget Management

- Monthly Budgets
- Category Limits
- CRUD Operations

Figure 5 - Backend API

- POST /auth/login
- GET /transactions
- POST /budgets

Figure 6 - MySQL Database

- users
- transactions
- budgets
- Data persisted successfully

End-to-End Test Scenarios

- Register a new user and receive JWT token.
- Login with valid credentials.
- Create income transaction and verify dashboard balance.
- Create expense transaction and verify remaining budget.
- Create monthly budget and validate duplicate prevention.
- View Budget vs Actual comparison.
- Verify transactions stored in MySQL database.
- Verify dashboard aggregates fetched from backend APIs.

Note: The UI and database figures above are illustrative placeholders suitable for documentation. Replace them with actual screenshots from your running application before final submission if required by the evaluator.